

Homework #2

Due date: 15th May 2026

1. Please submit your report (*.pdf) and schematic file (*.asc) to EECLASS [-5 pt]
2. Please name your report as hw2_studentID.pdf (e.g. hw2_113061000.pdf) [-5 pt]
3. Please name your schematic file as hw2_studentID.asc. If you submit additional files for specific problems, please name them as hw2_studentID_xxx.asc, where xxx indicates the problem name. For example: hw2_113061000_c.asc and hw2_113061000_d.asc. [-5 pt]
4. Please set white background and data trace with width = 3 in tutorial. [-5 pt]
5. You MUST extract the data of the waveform by cursor in tutorial. [-5 pt]
6. You MUST attach screenshot of the schematic with simulation commands and output waveform in your report. [-5 pt]
7. List calculation/design process clearly and mark proper information of all plots/figure required. [-5 pt]
8. Screenshots of the waveforms must include both x-axis and y-axis labels. [-5 pt]

Design of a BJT amplifier

Consider the Common-Emitter amplifier in Fig. 1, where $V_{CC} = 15V$, $R_{sig} = 2k\Omega$, $R_L = 10k\Omega$. Let 2N3904 be the NPN transistor used in the circuit. Please follow the steps below to **design R_1 , R_2 , R_C , and R_E** to achieve a voltage gain of $|v_o / v_{sig}| \geq 50$, an output voltage range of $-3V \leq v_o \leq 3V$ and a power consumption less than **150mW**.

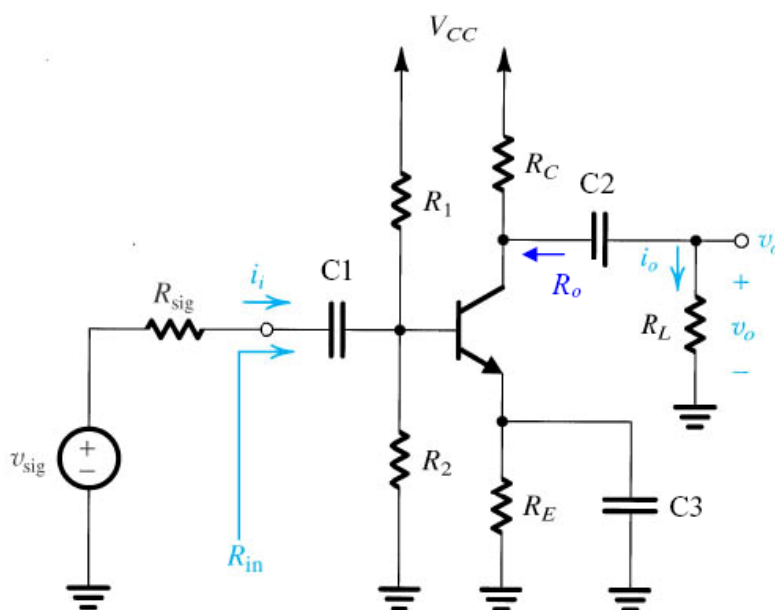


Fig. 1

- a. **Design the bias current I_C for the BJT.** Assume that $R_{in} \cong r_\pi$ and $R_o \cong R_C$ and let V_{RC} represent the voltage across R_C . Please prove the following equation in your report.

$$\frac{v_o}{v_{sig}} = - \frac{\beta I_C}{\beta V_T + I_C R_{sig}} \times \frac{V_{RC} R_L}{V_{RC} + R_L I_C} \quad (1)$$

According to the experience rule, set $V_{RC} = V_{CC}/3$. Use eq.(1) and the β vs. I_C characteristics in Table. 1 to **choose a desired bias current I_C** that gives $|v_o / v_{sig}| \geq 50$.

I_C (mA)	β_{dc}	V_{CE} (V)
0.5	303.714	2
10	296.53	2

Table. 1

- b. **Design R.** **Design all resistors** to bias the BJT with the desired bias current derived in (a). (Hint : use experience rules $V_{BB} \cong V_{CC}/3$, $I_{R1} = I_{R2} \cong 0.1 I_E$ and $V_T = 25mV$)
- c. **Simulation in LTspice.** Let v_{sig} be a 1kHz sine wave with an amplitude of 10mV. With the value of R designed in (b) and $C_1 = C_2 = C_3 = 100\mu F$, simulate in **LTspice** to show that the BJT amplifier has a gain greater than **50**. If the simulation result does not reach the requirement, fine tune the resistors to reach the requirement. Please use **transient** analysis and **plot v_o vs. time for 10ms** in your report. (**Hint: use the voltage source V_{SIN} for v_{sig} in LTspice**).
Also, write down all the resistor value and power dissipation (**the dc current supplied by $I(V_{CC}) * V(V_{CC})$**) in your report. (**Hint: use DC op pnt analysis to get $I(V_{CC})$**).
(Three screenshots are required:
1. one showing the schematic with the simulation commands,
 2. the other showing the waveform, including both the input signal v_{sig} and the output signal v_o , with the cursor indicating the output voltage. The waveform must include both x-axis and y-axis labels.
 3. Screenshot of the output of the DC operating point analysis.)
- d. Change the amplitude of v_{sig} to **100mV** and **plot v_o vs. time for 10ms** to show that the output range of v_o satisfies the required range (**$-3V \leq v_o \leq 3V$**). If the simulation result does not reach the requirement, fine tune the resistors to reach all the requirements. Write down the value of all resistors and the output range in your report. Does the mean value of v_o equal to zero? If not, please **explain** it in your report.
(Two screenshots are required:
1. one showing the schematic with the simulation commands,
 2. the other showing the waveform, including both the input signal V_{sig} and the output signal V_o , with the cursor indicating the output voltage. The waveform must include both x-axis and y-axis labels.)
- e. **Design the value of R_3** in Fig.3 such that the gain of the amplifier reduced by a **factor of 3**. Provide the **transient** simulation with LTspice to demonstrate your design is correct.

(Two screenshots are required:

1. one showing the schematic with the simulation commands,
2. the other showing the waveform, including both the input signal v_{sig} and the output signal v_o , with the cursor indicating the output voltage. The waveform must include both x-axis and y-axis labels.)

- f. Following your design in Fig.1, if R_L is reduced to 100Ω , please add an emitter follower stage (Fig. 4) between R_L and the CE amplifier to ensure that the voltage gain is still greater than 50 V/V . Please design the biasing current and R_B of the emitter follower and show your simulation result.

(Two screenshots are required:

1. one showing the schematic with the simulation commands,
2. the other showing the waveform, including both the input signal v_{sig} and the output signal v_o , with the cursor indicating the output voltage. The waveform must include both x-axis and y-axis labels.)

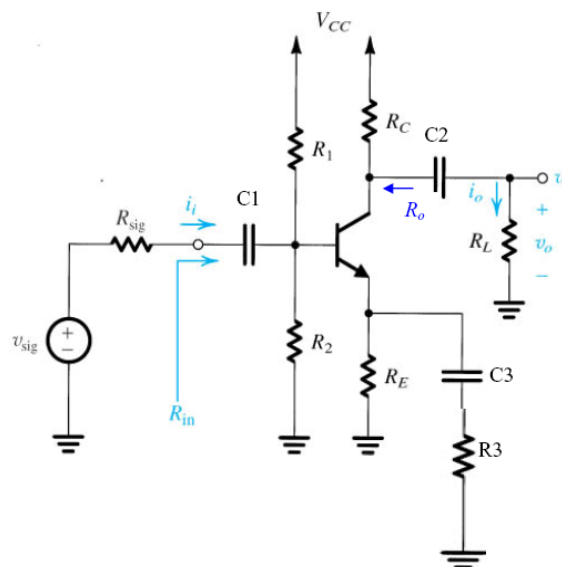


Fig. 3

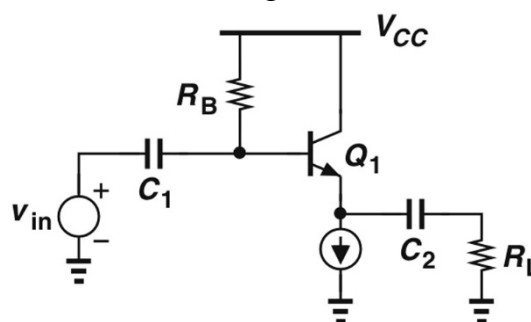


Fig. 4